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A hybrid renewable-based solution to electricity and freshwater problems in the off-grid Sundarbans region of India: Optimum sizing and socio-enviro-economic evaluation

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A R T I C L E I N F O

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A B S T R A C T

Stand-alone hybrid renewable energy systems have been proven as a promising pathway towards reliable and sustainable electrification of remote rural off-grid communities. The present study proposes such a system for a location near the Indian part of the Sundarbans, an area rich in biodiversity and endangered species. The hybrid system comprises solar photovoltaic modules, wind turbine, biomass generators with an electrolyzer-fuel cell-based storage system and can potentially replace the current kerosene-based arrangements, which have a significant damaging effect both on the surrounding ecosystem and the health of the residing population. The excess generation from the system is diverted to a reverse osmosis desalination system to meet the fresh water demands of the community. The sizing optimization of the hybrid configurations, performed in the MATLAB environment using a non-dominated sorting genetic algorithm-II, minimizes the cost of energy ($/kWh) and the damage to human health (DALYs) caused by the system while maintaining a demand supply reliability defined by the loss of power supply probability index. The optimized systems are analyzed based on crucial socio-enviro-economic indicators such as: cost of water ($\$/m^3$), lifecycle emission (kg CO_2-eq/yr), carbon emission penalty ($/kg), ecosystem damage (species·year), job creation, and human development index. The results indicate that the best configuration has an energy cost of 0.1967 $/kWh with a water cost of 0.86 $\$/m^3$. With the least carbon emission of 56,345 kg CO_2-eq/yr, 7.89E-2 DALYs of damage to human health, and 4.47E-4 species·year damage to the ecosystem, this system also imparts a positive effect on the community by creating 2.484 jobs and improving the living standards of the people with a human development index value of 0.5885. Moreover, the proposed hybrid system is able to mitigate 75,832 kg/yr of CO_2 emissions, preserving the ecological quality of the region.

1. Introduction

1.1. Background and literature review

In today's world, having access to clean water as well as affordable and sustainable energy is a basic requirement. However, global demand for freshwater and energy is increasing annually as a result of population growth, industrialization, and economic development. In countries such as India, despite government efforts, some remote areas still lack access to electricity or have inadequate access to electricity (Sarkar et al., 2019). There are numerous challenges in electrifying remote locations, such as long distances from the national grid, insufficient transmission and distribution infrastructural facilities, and difficult-to-reach terrains (Maisanam et al., 2021).

Renewable energy resources, which are abundant and environmentally friendly, have the potential to provide affordable, sustainable, and clean energy to address the problem of energy scarcity in remote areas and climate change. However, because of their intermittent availability, energy systems based on a single renewable energy source are not reliable (Hassan et al., 2022). The concept of a standalone hybrid renewable energy system (HRES) can be a promising alternative in remote areas, particularly on remote islands where grid connection is practically impossible. HRES uses multiple energy sources, mostly renewable energy resources, and sometimes in conjunction with a grid to meet a

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region's electrical power demand. HRESs offer the benefit of depending on multiple renewable resources to produce power, which overcomes the unreliability of a single renewable energy resource yet maintaining low greenhouse gas emissions (Baruah et al., 2021).

In a typical HRES, diesel generators are commonly used to improve the reliability of the system. However, there are several issues with diesel generators when they are used in HRES on remote areas. The primary challenges related with the use of diesel generators are fuel supply, transportation costs, and environmental pollution. In this context, fuel cells can be alternatively applied in HRES as an environmentally sustainable solution if the necessary hydrogen is produced through renewable sources (Rezaei et al., 2021). In addition, excess energy generated by the HRES can effectively be stored in a hydrogen tank through an electrolyzer, thus avoiding the deployment of battery storage, which has disposal issue. This is especially important in areas where the environment is endangered, such as the Sundarbans mangrove area in West Bengal, India. Installation of fossil-based power plants in this location may not be a good option, as traditional coal-based power plants generate a lot of solid waste, uncontrolled ashes, heavy metals, and other harmful materials (Flues et al., 2013). This could have a negative impact on the region's biodiversity. As a result, the installation of HRES appears to be a promising option in this region.

Hybrid system optimization, on the other hand, is a critical issue that will require a lot of thought and research effort. The first and most important step in this direction is to undertake comprehensive techno-economic studies in order to assess possible risks and trade-offs between design costs and energy production (Mytilinou and Kolios, 2019). Design and optimization of typical HRES is usually implemented using software tool packages such as HOMER (Homer Energy,USA, 2022), IHOGA/MHOGA (IHOGA/MHOGA, 2022), RETScreen (RETScreen, 2022), etc. or popular metaheuristic techniques such as genetic algorithm (GA) (Nagapurkar and Smith, 2019), Non-dominated Sorting Genetic Algorithm II (NSGA-II) (Deb et al., 2002), particle swarm optimization (PSO) (Shahid et al., 2022), many-objective population extremal optimization (MaOPEO) algorithm (Chen et al., 2019a), constrained multi-objective population extremal optimization (CMOPEO) algorithm (Chen et al., 2019b) etc.,

The effort has been made by various researchers for the advancement of HRES technology in order to provide efficient and cost-effective electricity for community scale applications. In a recent study, multi-criteria decision making (MCDM) based design optimization of an HRES integrating photovoltaic (PV), Wind Turbine (WT), Biogas generator (BG), Battery (BAT) and Hydro turbine (HT) was investigated by Ullah et al. (2021) for Ghulam Shah township, Pakistan. Gökçek (2018) investigated an HRES integrating PV, WT, BAT, diesel generator (DG), and reverse osmosis desalination (ROD) system in Bozcaada Island, Turkey, employing the HOMER Pro software to meet the electrical demand and produce freshwater. The system's levelized cost of water (LCOW) and levelized cost of electricity (LCOE) were reported to be \$2.20/m^3 and 0.308 \$/kWh, respectively. Kasaeian et al. (2019) investigated an Iranian rural electrification system comprised of PV module, BG, and DG and determined that the optimal LCOE was 0.193 \$/kWh. Das et al. (2022) used GA, PSO, and GA-PSO hybrid algorithms to optimize a hybrid PV/WT/BAT/DG/ROD system for supplying electricity and freshwater on Kutubdia Island, Bangladesh. Vishnupriyan et al. (2021) conducted techno-economic and multi-criteria decision analysis of renewable energy-powered ROD plants aimed at Indian cities. Murugaperumal and Vimal (2019) investigated the techno-economic feasibility of a bio-powered DG/WT/PV/BAT system for rural electrification in India. The hybrid system's optimal LCOE was estimated to be 10.18 Rs/kWh. Rajanna and Saini (2016) studied a HRES comprised of PV/WT/HT/BG/BAT through GA optimization technique for a remote area in India. The system provided LCOE of 0.089–0.108 \$/kWh along with excess energy production of 5.28–10.73%.

Recent research has identified social indicators as important parameters that should be addressed when modelling hybrid systems for long-term viability (Sawle et al., 2018). However, these indicators are often neglected, and only a few research publications have been found in which social indicators were taken into account. Khan et al. (2021) optimized a PV/WT/DG/BAT-based system using HOMER Pro software. They took into account social elements such as the human development index (HDI) and job creation (JC), as well as techno-economic and environmental parameters. Maqbool et al. (2020) investigated the social indicator JC as well as the economic metrics of a PV-Biomass-Grid-based optimum hybrid system based on locally accessible resources. Dufo-López et al. (2016) employed a multi-objective evolutionary algorithm to optimize a PV/WT/DG/BAT system in order to minimize net present cost while maximizing HDI and JC. In the HRES-based investigations, only a few studies considered the aspects of damage to human health and the ecosystem. Kiehbadroudinezhad et al. (2022) investigated the performance of a ROD plant integrated with a stand-alone HRES and estimated the damage to human health and the ecosystem caused by the diesel generator. A summary of recent HRES based works are presented in Table 1.

1.2. Bibliometric analysis

In addition to the aforementioned literature overview, the authors undertook a bibliometric analysis using VOSviewer software (VOSviewer, 2022) to provide a comprehensive and data-driven graphical representation of how scientific research in the field of HRES design works. It enables the authors to provide a perspective on the knowledge base and research trends in the examined field.

The Scopus database has been used to perform the bibliometric analysis on 2000 relevant original research articles published in peer-reviewed journals in the recent four years which were relevant to the scope of HRES. Fig. 1 shows the VOSviewer network visualization map of the keyword "hybrid renewable energy system" by the authors in their title, abstract, or keywords by co-occurrence cluster.

1.3. Observations and research contributions

According to the literature study and bibliometric analysis, the researchers' primary focus was on the techno-economic optimization of HRES. Social indicators are frequently overlooked in techno-economic analyses, with just a few research articles mentioning them. According to the preceding discussion and to the best of the authors' knowledge, optimization of a hybrid renewable energy system utilizing PV, WT, ROD plant with fuel cell-electrolyzer based storage facility aimed for electricity and freshwater production in an isolated place has not been investigated.

This paper proposes a novel integrated system that includes solar PV modules, a wind turbine, a biomass generator, and a ROD plant with a fuel cell-electrolyzer based storage facility for the remote off-grid Sundarbans region, West Bengal, India. The designed system is capable of providing electricity as well as fresh water for the region. The framework of the present study is depicted in Fig. 2. The main contributions of the present work are summarized below:

- An integrated hybrid energy system entailing solar PV module, wind turbine, biomass generator, and a ROD plant with a fuel cell-electrolyzer based storage facility is proposed and the system is comprehensively evaluated based on crucial socio-environmental and economic indicators.
- A detailed framework and energy management strategy have been developed to satisfy the freshwater and electricity demands, where water demand is mostly satisfied by utilizing the excess energy generated from the hybrid energy system.
- The study considers human health damage due to the GHG emissions from the hybrid energy system as one of the objective functions of the sizing optimization along with the cost of energy.

Table 1
Summary of recent HRES based works.

System structure	Software/Tools used	Location	Load Type	Consumption type	Performance criteria	Year	Ref
PV/WT/BAT/DG/ROD	HOMER, GA, PSO, GA-PSO hybrid algorithms	Kutubdia Island, Bangladesh	Remote island	Electrical, Freshwater	COE (0.234–0.265 $/kWh); NPC ($ 871,880-$ 988,540); EE (24,038–160,100 kWh/year)	2021	Das et al. (2022)
PV/FC/EL/DG/SC/BAT	HOMER	Khorfakkancity, UAE	Urban Community	Electrical	LCOE (0.346 $/kWh), RF (68.1%)	2021	Salameh et al. (2021)
PV/WT/BG/BAT/Grid	HOMER	Leopard Beach, China	Rural Community	Electrical	COE (0.201–0.256 $/kWh), EE (26.6–36.3%), NPC ($587,013-$748,607)	2020	Li et al. (2020)
PV/WT/BAT/DG/ROD	HOMER	Abo Ramad, Egypt	ROD plant	Freshwater	COE (0.164 $/kWh), NPC ($3.12 M)	2020	Elmaadawy et al. (2020)
PV/DG/BAT/ROD	HOMER	Sinai Peninsula, Egypt	ROD plant	Freshwater	COE (0.107 $/kWh), NPC ($502,662), RF (93.1%)	2020	Atallah et al. (2020)
PV/WT/DG/BAT/ROD	HOMER	Canary Islands, Spain	ROD plant	Freshwater	COE (0.404–0.478 $/kWh), RF (92–96%)	2019	Padrón et al. (2019)
PV/DG/BAT/ROD	HOMER	Khorasan, Iran	ROD plant	Electrical, Freshwater	LCOE (0.3975–0.5975 $/kWh), CDWP (1.59–2.39 $/m^3)	2018	Wu et al. (2018)
PV/WT/BAT	TRNSYS, jEPlus + EA	Hong Kong	Residential building	Electrical	LCOE (0.1251–0.5252 $/kWh)	2020	Liu et al. (2020)
PV/DG/BAT	LoadProGen,NGSA-II	Soroti, Uganda	Rural area	Electrical	LCOE, NPV, MIRR	2020	Fioriti et al. (2020)
PV/WT/BG/BAT	HOMER	UK and Bulgaria	Urban Communities	Electrical	LCOE (0.197–0.260 £/kWh)	2019	Tiwary et al. (2019)
PV/WT/PHS/BAT	PSO	China	Remote island	Electrical	COE (0.196–1.32 $/kWh) EE	2021	Javed et al. (2021)
PV/WT/BAT/PHS/MGT-CHP/Boiler	HOMER	Iran	Academic Institution	Electrical and Thermal	LCOE (0.09 $/kWh) NPC (M$ 42.5)	2021	Eisapour et al. (2021)
PV/WT/BAT	HOMER, MATLAB	Saudi Arabia	Residential house, Vehicle	Electrical	LCOE (0.4263–0.5702 $/kWh) NPC ($ 91,007-$ 133,882	2021	Al-buraiki and Al-sharafi (2021)
PV/WT/DG/BAT/Boiler/MGT-CHP/ROD	HOMER	Egypt	International airport	Electrical, Heat, and Freshwater	COE (0.0897–0.3683 $/kWh)	2021	Elkadeem et al. (2021)
PV/DG/WT/BAT	MATLAB	Maldives	Remote Island	Electrical	LCOE (0.12$/kWh)	2021	He et al. (2021)
PV/WT/BAT/DG	EnergyPlus, MATLAB, PSO,ArcGIS	Algeria	Residential building	Electrical	COE (0.21–0.26 $/kWh)	2021	Mokhtara et al. (2021)
PV/WT/DG/BAT	PSO, GOA, CSA	Yobe State, Nigeria	Remote community	Electrical	COE (0.3656–0.3674 $/kWh)	2020	Bukar et al. (2020)
PV/WT/BG/BAT	HOMER	Hassanabad,Iran	Rural Community	Electrical	COE (0.128 $/kWh) NPC ($904513)	2020	Jahangir and Cheraghi (2020)
PV/WT/BG/BAT	HOMER	Korkadu, India	Rural Community	Electrical	COE (12.99–13.72 Rs./kWh) NPC (1.18–1.26 Rs.in million)	2020	Murugaperumal et al. (2020)
PV/BAT/ROD/BWS	HOMER	Al Minya, Egypt	Isolated region	Electrical and Freshwater	COE (0.059$/kWh)	2019	Rezk et al. (2019)
PV/WT/BAT	HOMER	Crete Island, Greece	Seaport	Electrical	COE (0.08–0.241 $/kWh)	2021	Sifakis et al. (2021)
PV/WT/BG/Grid	HOMER	Kallar Kahar, Pakistan	Rural Community	Electrical	COE (0.0525–0.0713 $/kWh), NPC (M$ 166.35- M$ 197.40)	2018	Ahmad et al. (2018)
PV/BG/PHS/BAT	HOMER, WCA, MFO	India	Radio transmitter station	Electrical	LCOE (0.4864–0.4865 $/kWh)	2019	Das et al. (2019)
PV/WT/DG/BAT/EL	HOMER	Iran	Urban Community	Electrical, Heating, Hydrogen	COE (0.32–0.345 $/kWh)	2019	Akhtari and Baneshi (2019)
PV/WT/FC/EL	NSGA-II, CRITIC	China	Urban city	Electrical/ Hydrogen	LCOE (0.226 $/kWh) LPSP (4.01%) PAR (2.15%)	2020	Xu et al. (2020)
PV/WT/DG/FC/EL/BAT	HOMER	Turkey	Vacation homes and small market place	Electrical/ Hydrogen	COE (0.282–0.743 $/kWh)	2018	Duman and Güler (2018)
PV/WT/BAT/TLC	HOMER, GA	Australia	Community	Electrical/Heating	COE (0.226 $/kWh) NPC (1,243,383 $)	2021	Das et al. (2021a)

*BG: Biogas Generator, BWP: brackish water pumping, CRITIC: Criteria Importance Though Intercriteria Correlation, CSA:cuckoo search algorithm; EL: Electrolyzer, GOA: Grasshopper optimization algorithm; LPSP:Loss of power supply possibility, MIRR: Modified internal rate of return, MFO:Moth flame optimization, PAR: Power abandonment rate, PSO: particle swarm optimization,SC: Super capacitor, WCA:Water-cycle algorithm.

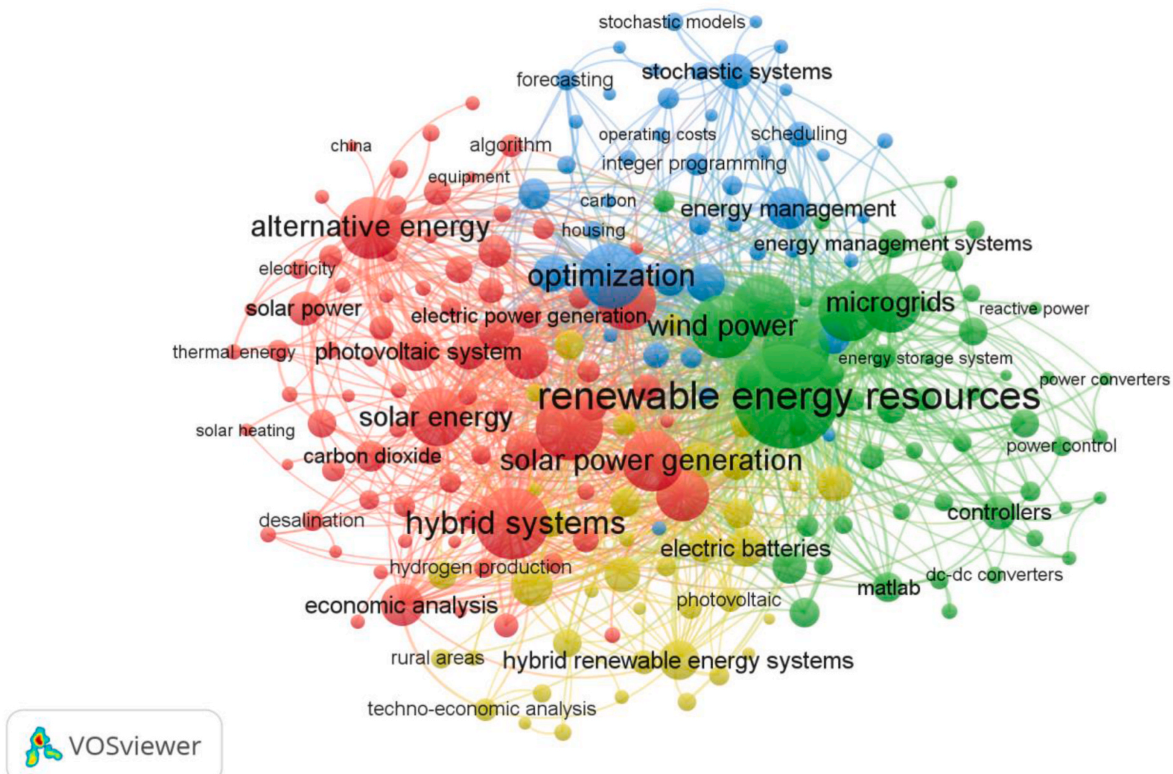

Fig. 1. VOSviewer network visualization map of the keyword "hybrid renewable energy system" by the authors in their title, abstract, or keywords by co-occurrence cluster.

- Key performance indicators such as reliability, lifecycle CO_2 emissions, job creation, problem associated with the ecosystem due to the GHG emissions, human development index, etc. have been extensively investigated.

2. Case study

2.1. Geographical and socio-economic information

The location selected for the analysis is Mousuni island, West Bengal, India (21.6624° N, 88.2023° E). The island is approximately 28 km^2 in size and it is a part of the Sundarbans region. More than 20% of the population of Mousuni in the Sundarban region is socially backward (Samanta et al., 2017). Approximately 68.15% of the population lives in poverty, with an income of less than $1.25 per day, and only 2.15% of individuals work in the organised sector (Samanta et al., 2017). People in Mousuni make their living mostly through farming and fishing. However, the tourism industry on Mousuni Island is flourishing due to its scenic beauty. Lately, it has become a popular tourist attraction, and many small-scale businesses with resorts and campgrounds have sprung up. Fig. 3 shows the location of the investigated area. This mangrove area and its ecosystems contribute numerous social, economic, and environmental benefits through fishing, tourism, wood, and non-wood products. However, its biodiversity and ecosystem are under threat due to numerous reasons including industrial pollution, overuse of forest resources, oil spillage, increased salinity level, and sea-level rise etc. (Paul, 2017). Therefore, it is paramount to establish a renewable-based energy system for the electrification of the study area.

2.2. Renewable resources

The selected study location is abundant in renewable energy sources. The solar irradiation and wind speed statistics for the study location were obtained from the NASA database using the HOMER software. The time series data of the solar irradiation of the targeted location is provided in Fig. 4.

Fig. 5 shows the wind speed data for the location. The annual average wind speed in the area is around 3.01 m/s. The study location is also abundant with local biomass.

2.3. Load assessments

Table 2 shows the daily load requirement of the community in the selected location. Here in this study, 250 households, 2 schools, streetlights, 1 primary health center, 20 small shops, 1 post office, and 2 business centers are considered. For the summer and winter seasons, the total daily community load demand is 1992.23 kWh/day and 1535.43 kWh/day, respectively.

The World Health Organization (WHO) identifies that around 20 L of freshwater per day is required for every person (Ibrahim et al., 2020). Therefore, for the 1500 people, approximately 30 m^3 of daily freshwater is needed for the study area. Moreover, a ROD system with the capacity of 1 m^3 of daily freshwater production requires 4.38 kWh electric energy (Ibrahim et al., 2020), hence the total electrical energy of 131.4 kWh/day is needed to desalinate for the study area. The monthly average load profile of the study is shown in Fig. 6.

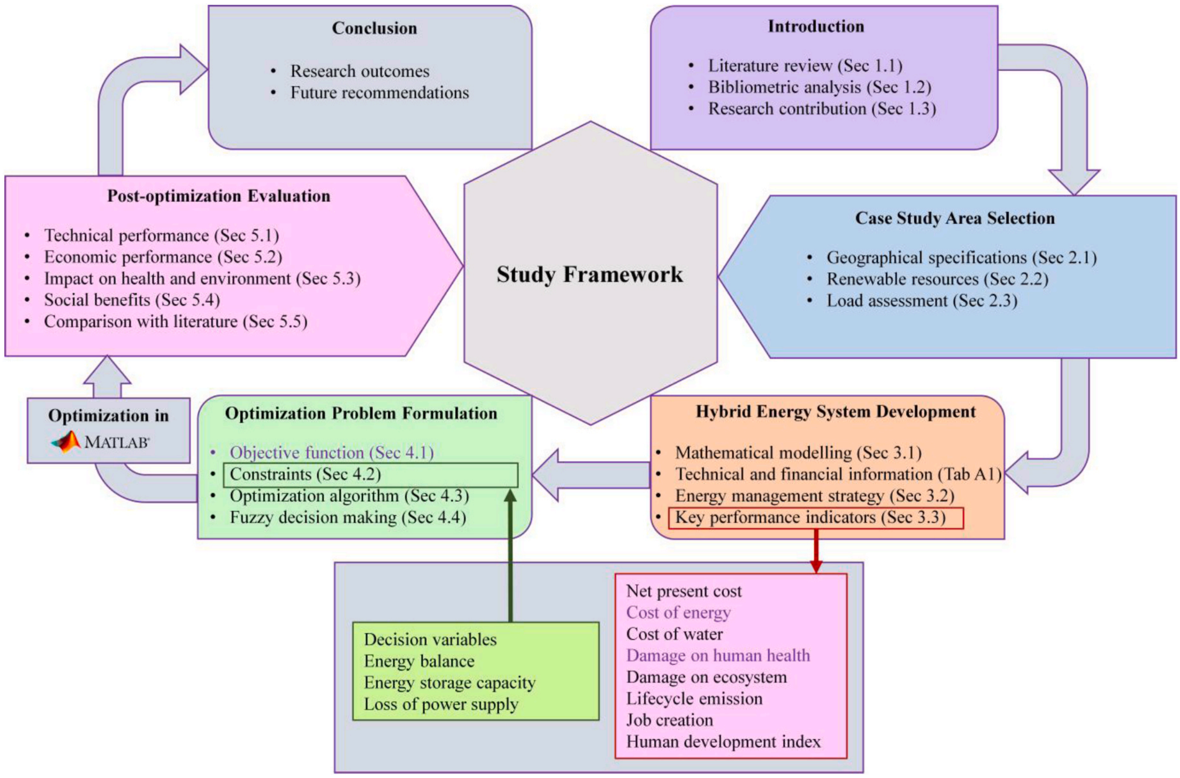

Fig. 2. Methodological framework of the present study.

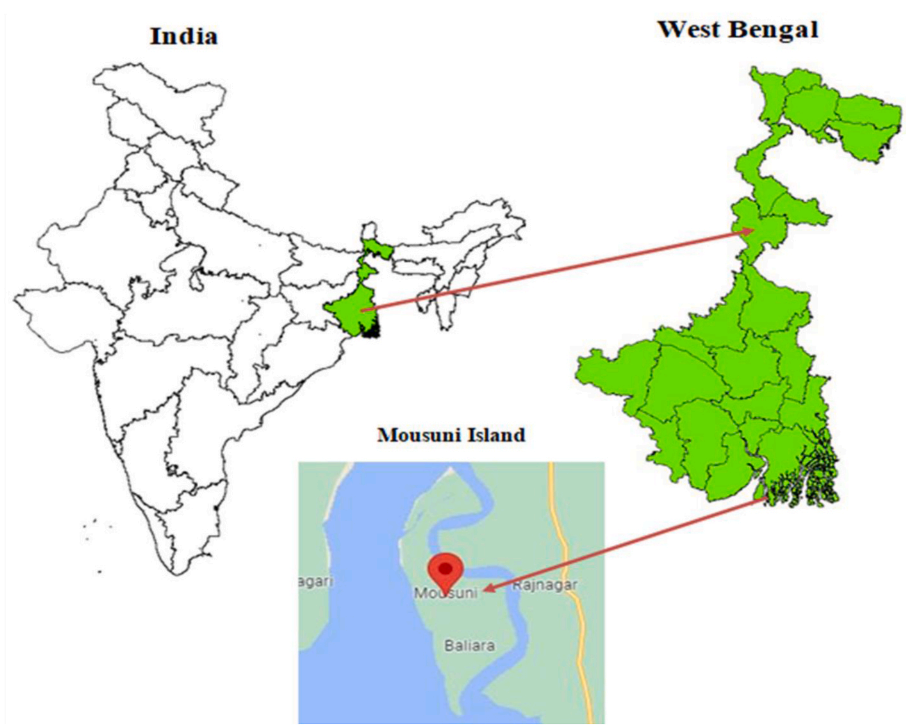

Fig. 3. Location of the investigated area.

3. Proposed hybrid system

The proposed hybrid renewable energy system is formed integrating solar photovoltaic modules and wind turbine generators as the primary energy sources with biomass generator as the backup. The system-generated excess energy is directed to a reverse osmosis desalination plant which caters to the community need of drinking water. Any surplus energy, after meeting the electricity demand of the community and the desalination system, is converted to hydrogen via an electrolyzer and stored. A proton exchange membrane fuel cell facilitates conversion of the stored hydrogen back to electrical energy to compensate for the deficiency in demands when that occurs. Energy excess to the storage

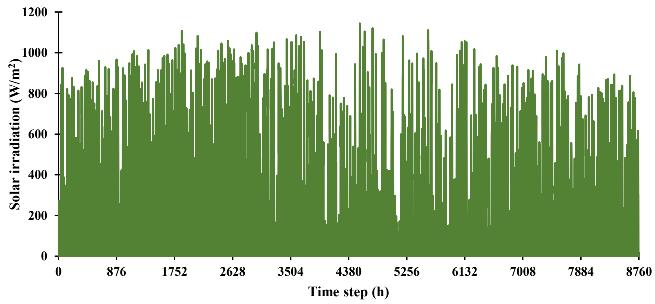

Fig. 4. Time series data for solar irradiation for the study area.

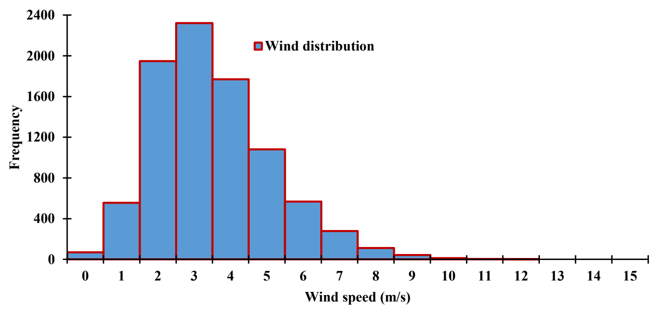

Fig. 5. Frequency of wind speed for the study area.

capacity is dumped via a resistance heater. A bidirectional inverter is exploited for the conversion of alternating current to direct current and vice-versa. A conceptual diagram of the proposed system is presented in Fig. 7.

3.1. Mathematical modelling

Wind Turbine: The output power generated by the wind turbine can be determined as follows (Chedid et al., 1998):

$$W_{WT} = \begin{Bmatrix} 0; \quad V \leq V_{cut,in} \\ a{*}V^3 - b{*}W_{rated}; \quad V_{cut,in} \leq V \leq V_{rated} \\ W_{rated}; \quad V_{rated} \leq V \leq V_{cut,off} \\ 0; \quad V > V_{cut,off} \end{Bmatrix} \tag{1}$$

where,

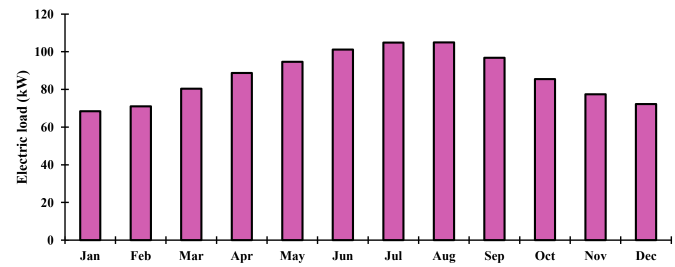

Fig. 6. Monthly average electric load data for the study area.

Table 2
Estimation of load requirement for the study area.

				Summer (Mar–Oct)		Winter (Nov–Feb)	
		Rating(W)	No. of use	Operation time (h)	Wh/day	Operation time (h)	Wh/day
(i)	House demand						
	Fan	70	2	7	980	0	0
	CFL bulb	25	4	7	700	8	800
	TV	100	1	6	600	6	600
	Refrigerator	150	1	24	3600	24	3600
	Mobile charger	6	2	3	36	3	36
	Water Pump	746	1	1	746	1	746
	Total (for 1 house)				6662		5782
Demand for 250 houses(kW/day)					1665.5		1445.5
(ii)	School demand						
	Fan	70	10	7	4900	0	0
	CFL bulb	25	15	8	3000	8	3000
	Computer	100	1	8	800	8	800
	Water Pump	746	1	1	746	1	746
	Total for 1 school				9446		4546
Demand for 2 schools(kWh/day)					18.892		9.092
(iii)	Primary Health Center						
	Fan	70	10	10	7000	0	0
	CFL bulb	25	20	12	6000	12	6000
	Computer	100	2	12	2400	12	2400
	Water Pump	746	1	3	2238	3	2238
	Refrigerator	150	2	24	7200	24	7200
	Total (for 1 centre)				24838		17838
Demand for 1center (kWh/day)					24.838		17.838
(iv)	Business center (2) + small shop (20) + Post office (1)						
	Fan	20	600	10	120000	0	0
	CFL bulb	25	600	7	105000	7	105000
	Computer	10	100	10	10000	10	10000
	Refrigerator	10	100	24	24000	24	24000
	TV	5	100	8	4000	8	4000
	Total (Wh/day)				263000		143000
Total (kWh/day)					263		143
(v)	Street lights						
	CFL bulb (Wh/day)	25	100	8	20000	8	20000
	Demand in kWh/day				20		20
Grand total load demand (i + ii + iii + iv + v) (kWh/day)					1992.23		1535.43

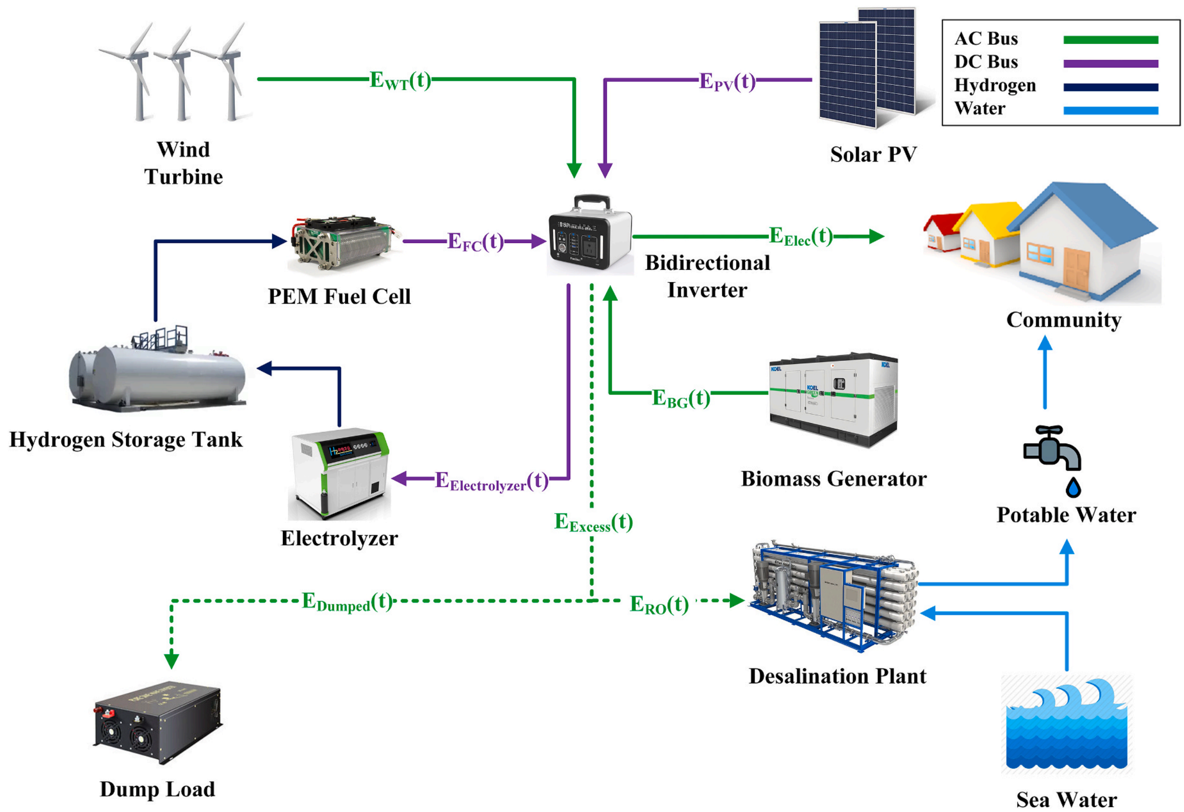

Fig. 7. Proposed hybrid energy system architecture for supplying electricity and water.

$$a = \frac{W_{rated}}{V_{rated}^3 - V_{cut,in}^3} \tag{2}$$

$$b = \frac{V_{cut,in}^3}{V_{rated}^3 - V_{cut,in}^3} \tag{3}$$

In the above equations, W_{rated} (kW) is related to the rated power, $V_{cut,in}$ (m/s) refers to the cut-in wind velocity, V_{rated} (m/s) corresponding to the rated wind velocity, and $V_{cut,off}$ (m/s) belongs to the cut-off wind velocity. The actual electrical power developed by the wind turbine can be estimated as follows (Chedid et al., 1998):

$$W_{e,WT} = W_{WT} * A_{sw} * \eta_w \tag{4}$$

where, total swept area and efficiency of wind turbine are denoted by A_{sw} and η_w, respectively.

PV module: The output power from the photovoltaic module can be determined as follows (Mousavi et al., 2021):

$$W_{PV} = Y_{PV} \times f_{PV}\left(\frac{I_T}{I_S}\right) \times [1 + \mu(T_c - T_s)] \tag{5}$$

where, Y_{PV} is the rated power of the PV module, f_{PV} is the derating factor, I_T is related to the solar irradiation incident, I_S is the solar irradiation at the standard test conditions, μ is the temperature coefficient, T_c is the cell temperature, and T_s is the cell temperature at standard conditions.

The cell temperature can be estimated as follows (Mandal et al., 2018):

$$T_c = T_a + I_T\left(\frac{T_{c,NOCT} - T_{a,NOCT}}{I_{T,NOCT}}\right) \times \left(1 - \frac{\eta_{PV}}{0.9}\right) \tag{6}$$

where, $T_{c,NOCT}$, $T_{a,NOCT}$, and $I_{T,NOCT}$ denote nominal operating cell temperature (NOCT), NOCT at the ambient temperature, and solar irradiation at NOCT, respectively and η_{PV} refers to the PV panel efficiency.

Inverter: A bidirectional inverter enables the conversion of power from DC to AC and vice versa. The output power $P_{out}(t)$ at the load side after conversion can be determined by the following equation (Das et al., 2021b):

$$P_{out}(t) = \eta_{inv} \times P_{in}(t) \tag{7}$$

where $P_{in}(t)$ and η_{inv} refers to the input power at any time and the inverter efficiency, respectively.

Electrolyzer: The electrical energy sent to electrolyzer at any time step can be determined as follows:

$$E_{Electrolyzer}(t) = [E_{PV}(t) + E_{WT}(t) - (E_{Elec}(t) + E_{RO}(t))] \times \eta_{inv} \tag{8}$$

The electrical energy is converted to hydrogen in electrolyzer and the hourly hydrogen generation, $H_{Electrolyzer}$ (kg), can be determined as follows (Ceylan and Devrim, 2021):

$$H_{Electrolyzer}(t) = 1.43 \times 10^{-3} + 2.39 \times 10^{-2} E_{Electrolyzer}(t) - 4.32 \times 10^{-5} E_{Electrolyzer}^2(t) \tag{9}$$

Fuel Cell: Fuel cell supplies necessary electrical energy during deficiency conditions. The energy requirement from FC is determined as follows:

$$E_{FC}(t) = \frac{[E_{Elec}(t) + E_{RO}(t) - (E_{PV}(t) + E_{WT}(t))]}{\eta_{inv}} \tag{10}$$

The quantity of hydrogen required to be converted by the FC, H_{FC} (kg), to supply the aforementioned energy deficit is determined by the following equation (Gharibi and Askarzadeh, 2019):

$$H_{FC}(t) = \frac{E_{FC}(t)}{\eta_{FC} \times HHV_{H_2}} \tag{11}$$

where, η_{FC} and HHV_{H_2} are fuel cell electrical efficiency (50%) and the higher heating value of hydrogen (37.8 kWh/kg), respectively (Gharibi and Askarzadeh, 2019).

Hydrogen storage: At any timestep when total production exceeds the demand, the hydrogen storage stores any surplus energy converted by the electrolyzer governed by the following equation:

$$H_{HS}(t) = H_{HS}(t-1) + H_{Electrolyzer}(t) \tag{12}$$

where, $H_{HS}(t)$ and $H_{HS}(t-1)$ are the total quantity (kg) of hydrogen stored in the storage at time step t and $(t-1)$.

At deficit conditions, stored hydrogen is converted to electrical energy. At such a condition, the hydrogen stored in the storage is determined as follows:

$$H_{HS}(t) = H_{HS}(t-1) - H_{FC}(t) \tag{13}$$

where, H_{FC} (kg) is the quantity of hydrogen taken from the storage and converted to electricity.

Biomass generator: Yearly electricity generator from biomass generator can be estimated as follows (Baruah et al., 2021):

$$E_{BG} = P_{BG} \times CF \times 8760 \tag{14}$$

where, P_{BG} and CF denote the rating of biomass generator and the capacity utilization factor, respectively.

The highest rating from the biomass generator can be estimated as follows:

$$P_{BG} = \frac{BG_{total} \times 1000 \times CV_{BG} \times \eta_{BG}}{365 \times 865 \times H} \tag{15}$$

where, BG_{total} is the total amount of available biomass, CV_{BG} denotes calorific value of the biomass, and η_{BG} is the efficiency of converting biomass to electricity and H represents hours of operation.

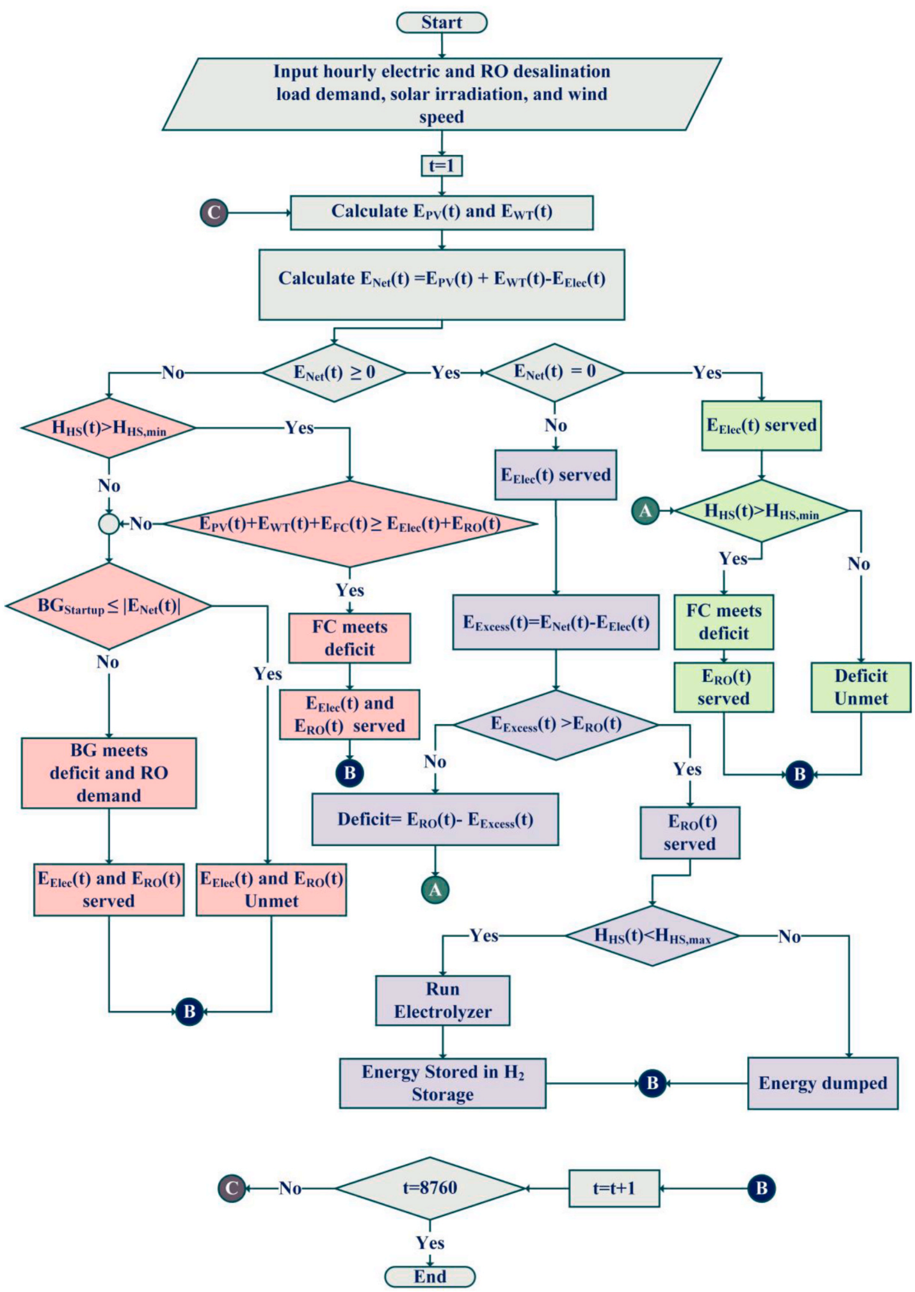

Fig. 8. Energy management strategy of the hybrid energy system for meeting electricity and water. Background colour notation: Red-$E_{Net}(t) < 0$, Purple- $E_{Net}(t) > 0$, Green- $E_{Net}(t) = 0$.

3.2. Energy management strategy

The control algorithm governing HRES operation is described in Fig. 8. Using the input parameters comprising load demands of electric and water along with renewables and temperature data, the control algorithm determines the demand-generation status and controls the storage strategies as follows:

- **Renewable energy generation is higher than the demand, $E_{Net}(t) > 0$:** In this scenario (indicated in purple background in Fig. 8), electric demand is fully served and some energy is found as excess ($E_{Excess}(t) = E_{Net}(t)$-$E_{Elec}(t)$). With excess energy sufficient to meet the entire RO demand ($E_{Excess}(t) > E_{RO}(t)$), the RO demand is met and rest of the surplus energy is used to produce hydrogen via electrolysis process and is stored in the H_2 tank at its highest capacity. For insufficient excess energy situation, the stored hydrogen is discharged and via fuel cell operation, rest of RO demand is met. In any case, when the highest H_2 storage capacity is reached and all the demand is met, the remaining excess energy is dumped via resistance heater.
- **Renewable energy generation is equal to demand, $E_{Net}(t) = 0$:** In the scenario (indicated in green background in Fig. 8), renewable production meets both the electrical and water demands where the later, in particular, is supported by the stored hydrogen via fuel cell.
- **Renewable energy generation is less than the demand, $E_{Net}(t) < 0$:** In this scenario (indicated in red background in Fig. 8), hydrogen fuel cell meets the shortage of electrical demand by discharging the H_2 tank. However, in any case when the deficit exists, the biomass gasifier delivers energy to satisfy demand given the condition of the load demand is the 30% of rated power or more. Any load demand below this threshold is termed as unmet load.

3.3. Indicators of HRES performance

Net Present Cost: The *NPC* can be calculated from Equation (16) (Kaabeche and Bakelli, 2019):

$$NPC = C_{cap} + C_{O\&M} + C_{rep} - C_{salvage} \tag{16}$$

The capital cost (C_{cap}), operation and maintenance cost ($C_{O\&M}$), replacement cost (C_{rep}), and the salvage value (C_{sal}) for each component can be determined using following equations.

$$C_{cap} = N_{comp} {}^{*} C_{cap,comp} \tag{17}$$

$$C_{rep} = N_{comp} {}^{*} C_{rep,comp} \sum_{j=1}^{N_{rep-comp}} \frac{1}{(1+i)^{L_{comp}*j}} \tag{18}$$

$$C_{O\&M} = N_{comp} {}^{*} C_{o\&m,comp} \sum_{k=1}^{L} \frac{1}{(1+i)^{k}} \tag{19}$$

$$C_{sal} = C_{rep,comp} {}^{*} \frac{R_{rem,comp}}{L_{comp}} \left(\frac{1}{(1+i)^{L_{comp}}} \right) \tag{20}$$

where N_{comp} is the number of optimal hybrid system components, and $C_{cap,\ comp}$ is the capital price of the component, $C_{rep,\ comp}$ is the replacement cost of the component at the termination of its lifetime, $N_{rep\text{-}comp}$ is the number of replacements of a component during its lifetime, $R_{rem,\ comp}$ is the remaining lifetime of the component, L_{comp} is the component lifetime, $C_{o\&m,comp}$ is the operation and maintenance cost of the component, and *k* is the index for project lifetime.

For the biomass gasifier generator, the cost of the fuel can be determined by Equation (21), whereby P_{fuel} is the fuel cost ($/L), and F_{fuel} is the annual fuel consumed (L/year).

$$C_{fuel} = P_{fuel} {}^{*} F_{fuel} \sum_{k=1}^{L} \frac{1}{(1+i)^{k}} \tag{21}$$

The cost associated with the RO can be obtained as follows (Maleki, 2018):

$$TC_{RO} = C_{RO} + O\&M_{RO} + C_{WTa} + C_{MR} + C_{Chem} \tag{22}$$

where C_{RO}, $O\&M_{RO}$, C_{MR}, and C_{Chem} are the capital cost, O&M cost, membrane replacement cost, chemical cost of the RO system, and C_{WTa} is the capital cost of the freshwater tank, respectively.

Cost of Energy: The COE is the indicator of the economic viability of a hybridized project and is determined from Equation (23), whereby *NPC* is the total net present cost of the hybrid components, E_s is the annual load served, and $CRF\,(i, L)$ is the capital recovery factor.

$$Min\ COE = \frac{NPC \times CRF(i,L)}{\sum_{t=1}^{8760} E_S} \tag{23}$$

The value of $CRF(i,L)$ depends on the real annual interest rate (i) and the project lifetime (L), and is determined by Equation (24).

$$CRF(i,L) = \frac{i(1+i)^{L}}{(1+i)^{L}-1} \tag{24}$$

The cost and technical details of all the components of the HRES have been reported in Appendix (Table A1).

Lifecycle emission: The life cycle emissions over the year of the HRES can be determined by the following Equation (25) (Das et al., 2017).

$$LCE = \sum_{m} \sum_{i=1}^{T} \psi_m E_{m,i}\ ,\ m \in \{PV, WT, BG, FC, Electrolyzer, INV,\ ROD\} \tag{25}$$

where, ψ refers to the lifetime equivalent CO_2 emissions and *E* refers to the amount of energy stored (via FC and electrolyzer) or converted (by PV module, wind turbine, biomass generator, and inverter) while meeting the load demand over the period of T (8760 h).

Damage to human health and ecosystem: The CO_2 emission caused by the system components will have certain contribution in elevating the global mean temperature, the effect of which will cause adverse effect on the ecosystem and human health (Shi et al., 2022). The damage to human health (HHD) is quantified by disability adjusted life years (DALYs), which represents the years of life lost due to premature mortality, or years of healthy life lost due to disability (WHO, 2013). The unit for measuring ecosystem damage (ESD) is species·yr which is defined as the total loss of local species integrated over a year. In this study, the human health and ecosystem damage factors have been considered as 1.40E-06 DALY/kg CO_2-eq and 7.93E-09 species·yr/kg CO_2-eq (Huijbregts et al., 2017),respectively.

Human development index (HDI): Human development index (HDI) represents the socioeconomic development of people in general. It may also be seen in the energy usage patterns of communities in a certain area (Sawle et al., 2018). Here, it is considered that the annual excess energy can be used by local small enterprises, which may improve the standard of life and hence the HDI. The HDI can be estimated by the following relation (Maqbool et al., 2020).

$$HDI = 0.0978\ \ln\left[\frac{\left(E_{load} + min\left(f_{ex,max} \cdot E_{ex}\ , f_{load,max} \cdot E_{load}\right)\right)}{Number\ of\ people}\right] - 0.0319 \tag{26}$$

where, E_{load} and E_{ex} represent annual AC load and annual excess energy of the system. The values of factors $f_{ex,max}$ and $f_{load,max}$ are considered as 0.2 and 0.5, respectively (Maqbool et al., 2020).

Job creation (JC) factor: The job creation (JC) factor refers to the process of creating new jobs, which aids in the reduction of unemployment in a certain location. The job creation (JC) factor is estimated by the following relation.

$$JC = JC_{PV} \times P_{PV} + JC_{WT} \times P_{WT} + JC_{BG} \times P_{BG} + JC_{FC} \times P_{FC} + JC_{ROD} \times D_{ROD} \tag{27}$$

where, JC_{PV}, JC_{WT}, JC_{BG} JC_{FC}, and JC_{ROD} are the number of jobs per MW of PV energy, number of jobs per MW of the wind turbines, number of jobs per MW of the biomass generator, number of jobs per MW of the fuel cells and jobs created by ROD plant per m^3 of water demand, respectively. The major input parameters required for estimating JC factor are provided at Table 3.

4. Optimization framework

4.1. Objective functions

The study considers minimization of the COE and the human health damage (HHD) while supplying both electric and fresh water demands under specific load reliability (LPSP $= 1 \pm 0.5$) in the development of optimization as follows:

$$F = min\{COE,\ HHD\} \tag{28}$$

The COE is a widely used economic indicator to determine the optimal sizing of hybrid energy system configurations with greater transparency (Elkadeem et al., 2021) and has received pronounced acceptance amongst the investors. On the other hand, lifecycle GHG emissions is an important parameter for determining the environmental suitability of the hybrid energy system implementation leading to an overall sustainable development. However, quantification of GHG emissions alone cannot adequately reflect the consequences for the common people, particularly those living in the rural areas. Therefore, the study considers minimizing human health damage as one the objective functions of optimizing the proposed hybrid renewable systems to increase the acceptance of these system amongst general public by addressing the concern of their well-being. Such consideration will also help raise social awareness about the environmental soundness and health implications of the present means of utilities.

4.2. Constraints

Range of decision variables: The decision variables are the number of system components for optimizing the size of the HRES. In the process of optimization, the solution space is subjected to predefined upper and lower ranges to achieve the efficient computation as follows:

$$N_{m,min} \le N_m \le N_{m,max}\ ,\ m \in (\ PV, WT,\ BG,\ FC,\ Inv,\ H_2\ Tank, Electrolyzer) \tag{29}$$

where N_m represents the number of a system component of m, and $N_{m,min}$ and $N_{m,max}$ refer to the minimum and maximum number of system component of m, respectively. The upper and lower bounds are used to limit the solution space to reduce the processing time. This is determined at an early stage of optimization after running the process on a trial-and-error basis.

Table 3
Major input parameters for JC factor estimation (Das et al., 2022; Dufo-López et al., 2016; Sawle et al., 2018).

Components	Factors
PV module	2.7 jobs/MW
Wind turbine	1.1 jobs/MW
Biomass power generation	1.64 jobs/MW
Inverter	0
Fuel cell	1.2 jobs/MW
ROD plant	0.056 jobs/m^3

Energy balance constraints: The total energy generation from all the system components in an hour must be equal or greater than the hourly load demand.

$$E_{PV}(t) + E_{WT}(t) + E_{BG}(t) + E_{FC}(t) - E_{Electrolyzer}(t) = E_L(t) + E_{Excess}(t) \tag{30}$$

E_{PV} E_{WT}, and E_{BG} are the energy generation from PV, WT, and BG where,E_{FC} and $E_{Electrolyzer}$ are the electrical energy supplied and stored while meeting the total electrical demand $E_L(t)$ at that hour.

Energy storage capacity constraints: The hydrogen storage capacity in the system is governed by the following constraint:

$$H_{HS,min} < H_{HS}(t) < H_{HS,max} \tag{31}$$

where, $H_{HS,min}$ and $H_{HS,max}$ are the minimum and maximum capacity (kg) of the hydrogen energy storage.

Reliability constraints: The reliability of the HRES is assessed using the loss of power supply probability (LPSP), a widely accepted reliability index, which refers to the probability of the system's failure to satisfy the load demand over a period of one year. The LPSP is determined using Equation (32)(Das et al., 2017).

$$LPSP = \frac{\sum_{t=1}^{T} LPSP(t)}{\sum_{t=1}^{T} E_L(t)} \tag{32}$$

LPSP is calculated at the worst-case scenario of power supply when the renewable, supplementary and the storage systems fail to generate or supply any power to meet the demand. It can be found over the period of T (8760 h) using Equation (33).

$$\sum_{t=1}^{T} LPSP(t) = \sum_{t=1}^{T} \left(E_L(t) - \left[E_{Gen}(t) + E_{FC,max}(t)\right] \times \eta_{inv}\right) \tag{33}$$

The system reliability constraint is then expressed as:

$$LPSP \le LPSP_{Desired} \tag{34}$$

where, $LPSP_{Desired}$ refers to the required reliability of the system and is considered as $1 \pm 0.5\%$ for this study.

4.3. Optimization algorithm

NSGA-II is the multi-objective variant of the genetic algorithm, a widely known and utilized evolutionary algorithm, which features an elitist strategy that allows more rapid sorting of the non-dominated solutions than most of its counterparts. Along with better convergence, NSGA-II maintains a diverse spread of solutions on the Pareto-optimal front. The multi-objective GA has higher effectiveness in finding global optima of objective functions than other types of optimization tools like PSO along with the capability of handling larger numbers of parameters (Erdinc and Uzunoglu, 2012). The details of the algorithm can be found in (Deb et al., 2002). In this research, a sensitivity analysis has been carried out to determine the different optimization parameters and these are adopted according to the outcomes as illustrated in Table 4.

4.4. Determination of final solution

A set of non-dominant solution on the Pareto front is found from the multi-objective solution, therefore, a decision-making technique is

Table 4
NSGA-II optimization parameters.

Optimization Parameters	Value
Population size	500
Maximum generation	500
Crossover rate (SBX crossover)	0.9
Mutation rate (Poly mutation)	0.1

required to find the best possible solution (trade-off) on the Pareto front. In this study, fuzzy decision-making process is used to choose the optimal solution from the Pareto front. The fuzzy membership value of the jth objective function is calculated as follows (Biswas et al., 2018), whereby F_j^{max} and F_j^{min} is the maximum and minimum fitness value of the jth objective function.

$$\chi_j = \begin{cases} 1 & for\ F_j \le F_j^{min} \\ \dfrac{F_j^{max} - F_j}{F_j^{max} - F_j^{min}} & for\ F_j^{min} < F_j < F_j^{max} \\ 0 & for\ F_j \ge F_j^{max} \end{cases} \tag{35}$$

The normalized membership function, χ^k for each non-dominant solution is defined as (Brka et al., 2015):

$$\chi^k = \frac{\sum_{i=1}^{N_{obj}} \chi_j^k}{\sum_{k=1}^{M} \sum_{j=1}^{N_{obj}} \chi_j^k} \tag{36}$$

where M and N_{obj} are the non-dominated solutions and the objective function number, respectively and final solution is achieved from the normalized membership functions with the maximum χ^k value.

5. Results and discussion

This study considers optimization of hybrid renewable systems comprising PV, WT, BG, FC, electrolyzer, and hydrogen storage for simultaneous supply of electricity and potable water to a remote area adjacent to the Sundarbans. To understand the effect of hybridization on the KPIs, three different hybrid scenarios have been studied. The optimized outcomes of these systems are summarized in Table 5. A multi-objective genetic algorithm or NSGA-II gives a well-defined Pareto optimal solution as reported in Fig. 9 and final solution is determined using fuzzy decision-making process.

5.1. Technical performance

5.1.1. System sizing and energy balance

Fig. 10 represents the energy interaction scenario for the three hybrid systems in a typical day. As it can be observed, middle part of the day exhibits high volume of renewable production exceeding the demand, the larger share of which comes from PV due to the higher solar potential of the area. BG-integrated systems receive a substantial contribution from this non-renewable source as it covers the peak demands incurred throughout the night and the earlier parts of the day when there is little or, no renewable production. This is also reflected in the annual energy production scenario (Table 5) as BG shares around 50–55% of total production in these two systems (PV/WT/BG/FC and PV/BG/FC). Exclusion of BG from the system (PV/WT/FC) radically increases the size of all the components, leading to a subsequent jump in the excess energy production as evident from Fig. 10 (b). For all the systems, recourse to FC can be observed especially for meeting RO demands at night-time, at the later part of the day with a demand surpassing renewable production, and when the BG's start-up threshold is more than the demand itself. With no BG to back-up, night-time energy supply become FC-oriented, leading to a large overall increase in the size of this component as well. Absence of wind turbines in the system causes

Table 5
Summary results of the optimized HRES.

Parameters	PV/WT/BG/FC	PV/BG/FC	PV/WT/FC
PV module (kW)	170	300	811
Wind turbine (kW)	90	–	910
Biomass generator (kW)	150	150	–
FC (kW)	30	38	226
H_2 tank (kg)	24	36	1068
H_2 Electrolyzer (kW)	42	120	264
Converter (kW)	130	150	815
PV energy (kWh/yr)	264,956	467,240	1,263,637
Wind energy (kWh/yr)	152,176	–	1,538,671
Biomass energy (kWh/yr)	506,112	462,202	–
Excess energy (kWh/yr)	49,960	14,061	1,392,700
Total Electric Demand (kWh/yr)	803,547	803,547	803,547
Total RO demand (kWh/yr)	39,967	39,967	39,967
Unmet electric demand (kWh)	4919	7518	8252
Unmet RO demand (kWh)	625	1123	323

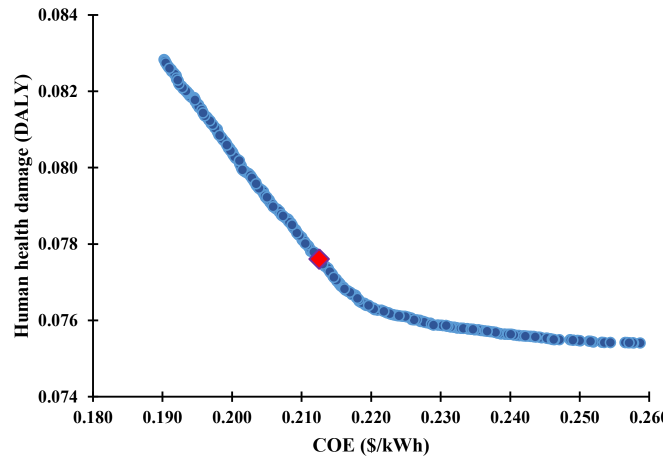

Fig. 9. Pareto front for PV/WT/BG/FC-based HRES. The red box indicates optimal solution using Fuzzy decision-making process.

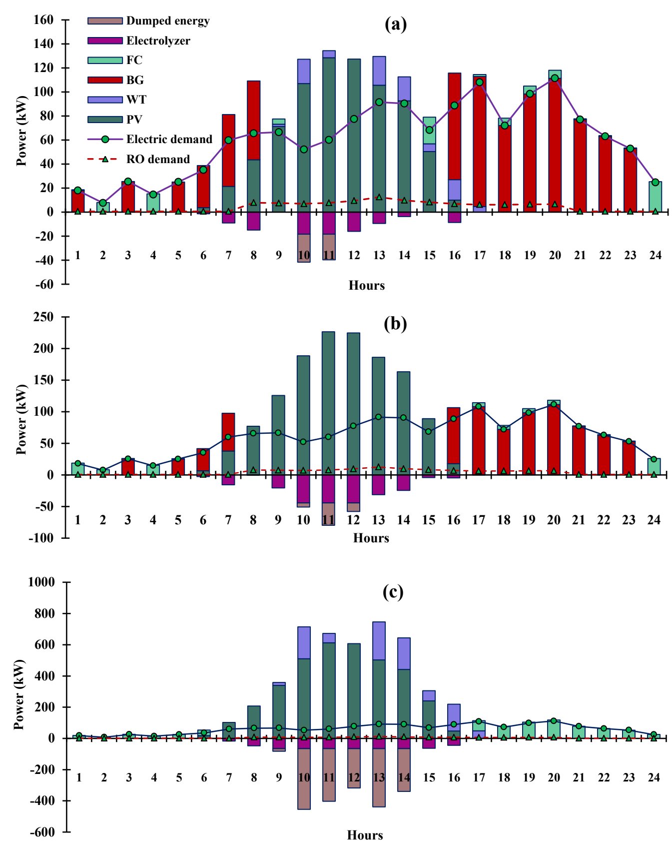

Fig. 10. Power interaction scenario for (a) PV/WT/BG/FC (b) PV/BG/FC and (c) PV/WT/FC system in a typical day. Positive Y-axis indicates production and supply while negative axis direction indicates storage and disposal of additional production.

around 43.33% increase in the PV size with unchanged BG size to meet the demand, but reduces excess energy production by 72%. On the other hand, integrating wind turbines scales down the sizes of FC, H_2 tank, and electrolyzer by 21.05%, 33.33% and 65%, respectively.

5.1.2. Energy system efficiency analysis

The energy system efficiency identifies the effectiveness of the HRES. It is defined as the energy supplied to meet the demand to the energy generation from the HRES. Results as reported in Table 6 reveal that both PV/WT/BG/FC and PV/BG/FC have comparable but relatively higher energy system efficiency. The energy loss occurs due the AC-DC conversion and the loss associated with the FC energy conversion process. The PV/BG/FC-based HRES has higher loss than the PV/WT/BG/FC one due to higher contribution of the PV module for the former than the later one. However, the excess energy for PV/WT/BG/FC-based HRES are larger (49,960 kWh/yr) than the PV/BG/FC (14,061 kWh/yr). This is because of the higher PV and wind energy share for meeting load demand in the PV/WT/BG/FC-based HRES than the PV/BG/FC one. Without backup generators in the system, the variable load demand leads to oversizing of the system components, which in turn results in a substantial quantity of excess production. Moreover, absence of AC backup system, the AC-DC conversion and FC energy conversion requirement significantly increases. The higher loss associated with such frequent conversion and the over-production cumulatively cause a considerable reduction in the overall system efficiency, as it is observed for PV/WT/FC system.

5.2. Cost analysis

Fig. 11 represents component-wise share in the total net present cost of the three hybrid systems. For the BG-integrated systems, one of the largest cost shares is associated with biomass generator and the corresponding fuel costs. For PV/WT/BG/FC system this two expense covers respectively 32.84% and 22.38% of the total expenses, while for PV/BG/FC it is 30.19% and 20.35%, respectively. In these two systems, aggregated cost of renewable generators is comparable. However, the difference in electrolyzer cost marks the difference between these two systems as in PV/BG/FC system this increases by 185%. Higher size of electrolyzer causes this cost hike, which also attributes to the reduced quantity of dumped energy from this system. Exclusion of BG makes the system cost-intensive as the size of each of the components increases radically. Around 48% of this cost is due to the wind turbine generators. Fig. 12 depicts the cost comparison for electricity and water from these systems. As it can be observed, the least energy cost (0.1967 $/kWh) is offered by the PV/WT/FC/BG system with a water cost of 0.86 $/m^3, also the lowest of those offered by the other systems considered.

5.3. Impact on human health and environment

This study quantifies the impact of the hybrid energy systems on environment by the emission of equivalent CO_2 throughout the lifecycle by each of the components. Moreover, how such emissions affect the human health and the ecosystem of the study area are also taken into account. As presented in Fig. 13, PV/WT/BG/FC causes the least emission of the three systems and consequently the estimated damaging effect on human health and ecosystem is also the least. In the yearly total of 56,345 kg of CO_2 emission, about 54% is from BG and the least (3%) is

Table 6
Energy system efficiency for the different HRESs.

HRES	Energy served (kWh/yr)	Total energy generation (kWh/yr)	Excess energy (kWh/yr)	Energy loss (kWh/yr)	Energy system efficiency (%)
PV/WT/BG/FC	797,596	923,244	49,960	75,688	86.39
PV/BG/FC	795,512	929,442	14,061	119,869	85.59
PV/WT/FC	795,295	2,802,308	1,392,700	614,313	28.38

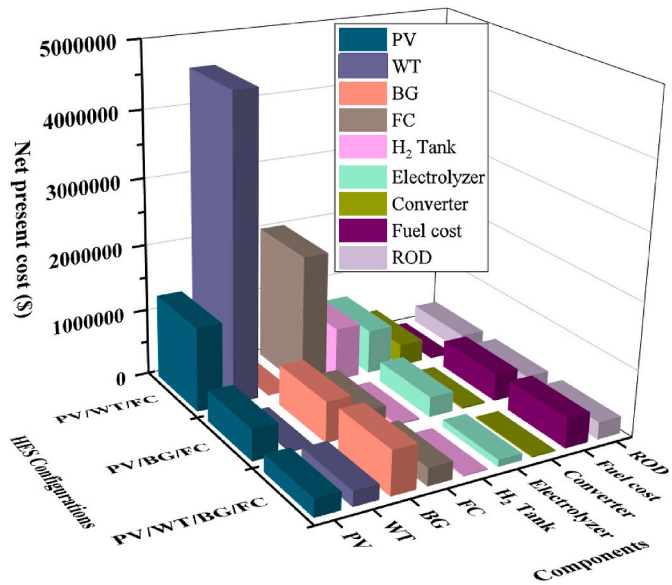

Fig. 11. Breakdown of total net present cost of the three systems.

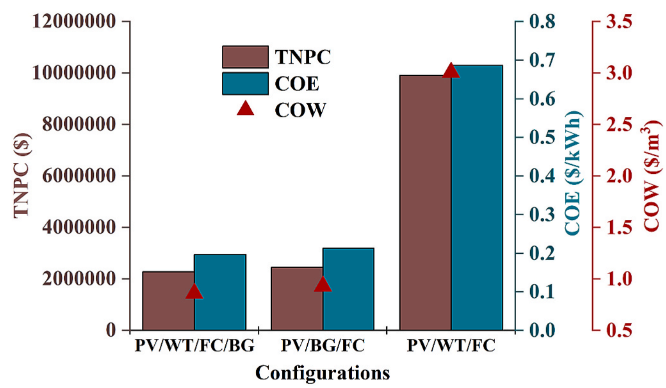

Fig. 12. Net present system cost, cost of energy, and cost of water of the three systems.

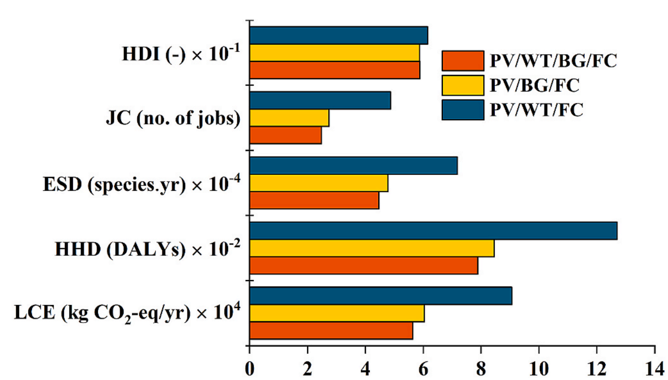

Fig. 13. Impact on human health, ecosystem, and social benefit creation by the different HRESs.

caused by wind turbines, as it can be observed from Fig. 14. Without wind turbine in the system, BG emission drops down by 8.67%, but higher PV size and corresponding higher emission factor of PV modules causes around 76% increase in the emission of this component itself and results in 7% increase in the overall emission. Sizeable hardware components considerably increase the total emission from PV/WT/FC-based system, where the emission from PV alone is found to be greater than the entire emission from the least carbon-emitting system.

5.3.1. Carbon emission penalty

In this study, the CO_2 emission penalty has been assessed by the lifecycle emission of HRES components. Renewable sources generate no operational CO_2 emission during their operating process; however, the HRES have the LCE, which is the equivalent CO_2 (kg CO_2-eq/kWh). The LCE incorporates equivalent CO_2 emissions from the HRES used to manufacture, transport, and recycle the system components. Results show that the PV/WT/BG/FC-based ROD system generates 56345 kg of CO_2-eq/yr. The study considers the carbon tax is 0.015$/kg of CO_2 emission in association with the literature (Maisanam et al., 2021). The annual cost due to the emission is found to be $845. In addition, carbon penalty costs for the PV/BG/FC and PV/WT/FC-based ROD system share $906 and $1359, respectively. The COEs including the carbon penalty are 0.1987 $/kWh, 0.2146 $/kWh, and 0.6898 $/kWh, respectively for the PV/WT/BG/FC, PV/BG/FC and PV/WT/FC-based HRES systems.

5.3.2. Estimation of fuel savings and CO_2 mitigation

The study estimates kerosene savings, which is currently being used for household lighting in the study area. Substantial benefits would be achievable by switching the kerosene lanterns to the electrification with the proposed HRES. It is estimated that the specific kerosene consumption for a typical lantern is 0.04 l/h and each lantern can provide lighting for around 4 h (Chaurey and Kandpal, 2009). Each household uses two lanterns or wick lamps for lighting while business center uses one lantern. The total kerosene consumption of the area is, therefore, around 30,950 l/yr. The CO_2 emission coefficient for kerosene is 2.45 kg CO_2/l (Chaurey and Kandpal, 2009). Therefore, the proposed HRES system is likely to mitigate the CO_2 of 75,832 kg/yr when installed.

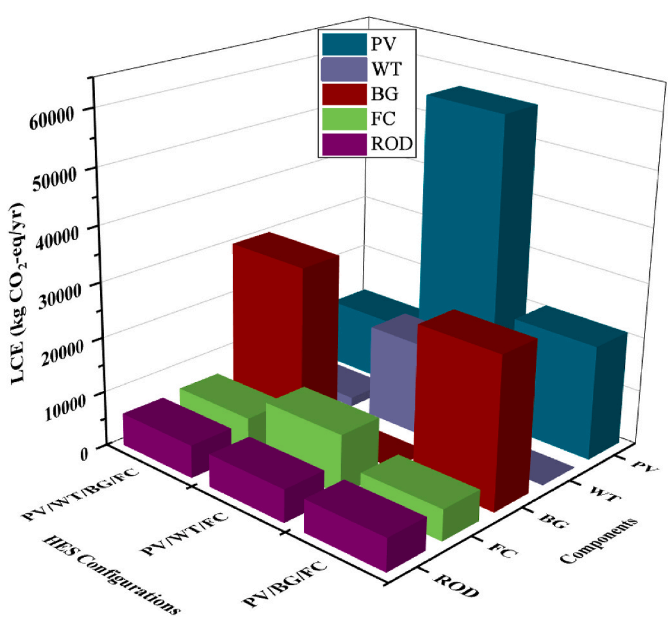

Fig. 14. LCE contributions from the different components of the hybrid systems.

5.4. Social benefits

In this study, two important social benefit factors, viz. HDI and JC are estimated for the optimized conditions. Excess energy from the energy system has the ability to improve the people's standard of living in the studied area. As reported in Fig. 13, the HDI values for the PV/WT/BG/FC, PV/BG/FC and PV/WT/FC configurations are estimated to be 0.5885, 0.5877 and 0.6153, respectively. The higher values of JC factor denote the higher potential of creating jobs. The PV/WT/FC configuration has the highest JC factor (4.871), followed by PV/BG/FC (2.736) and PV/WT/BG/FC (2.484). With the electrification based on the hybrid renewable system, the basic lighting of the community will improve replacing the prevailing kerosene lanterns. Such improvement will enhance the home study hour of the children reducing eye-strain and other associated health damage (Sharma et al., 2019). Availability of better lighting will increase the participation in study as well which will potentially contribute to improving the current rate of literacy. The business situation in the community will also upgrade with longer night-time lighting hours made possible by the reliable supply of electricity by the proposed system.

5.5. Comparison with literature

Amongst the three optimized systems, PV/WT/FC/BG-based offers the least energy cost of 0.1967 $/kWh with 56,345 kg CO_2-eq/yr of lifecycle emission. The economic and environmental performance of this system has been further validated by the published literature and a comparative depiction of energy costs amongst the studies is presented in Fig. 15. The study carried out by Das et al. (2022) offered the least cost of 0.234 $/kWh with 40,239 kg CO_2-eq/yr of emission. Salameh et al. (2021) optimized PV/FC/DG/FCESS-based system which offered energy cost of 0.341$/kWh and emitted 3,844,051 kg of CO_2 per year. A PV/FC based system studied by Peláez-Peláez et al. (2021) fetched an energy cost of 0.8399 $/kWh. The optimal PV/WT/FC-based proposed by Xu et al. (2020) offered an energy cost of 0.226$/kWh. The optimization study of Duman and Güler (2018) found a PV/WT/FC/DG-based system with an energy cost of 0.282 $/kWh. The system proposed by the present study offers significantly better and competitive economic and environmental outcomes.

6. Conclusions

The study examines the performance of a fuel cell storage-based hybrid energy system for meeting electric and freshwater demands of a remote off-grid area near the Sundarbans region in West Bengal, India. A methodological outline has been developed so as to utilise the excess energy to meet the freshwater demand. This research analysis considers

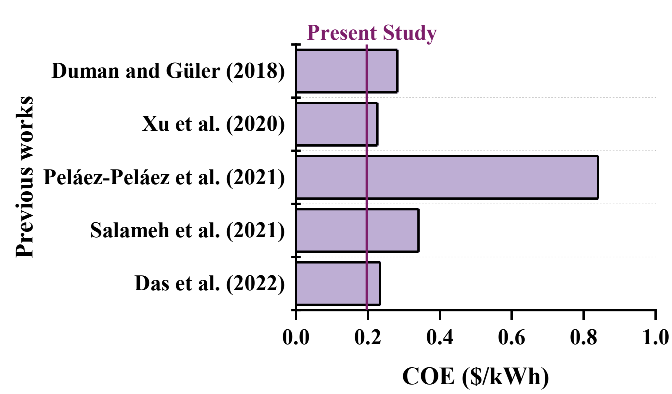

Fig. 15. Comparison of present study with previous works. The vertical line represents the energy cost offered by the optimum HRES (PV/WT/FC/BG) proposed in the present study.

the cost of energy and human health damage due to GHG emissions as the objective functions to determine the optimal component sizing while satisfying allowable reliability criteria. Several key performance indicators have also been evaluated during the optimization process. The relevant outcomes are listed as follows:

- The lowest energy cost (0.1967 $/kWh) is offered by the PV/WT/BG/FC system, with the least water cost (0.86 $/m^3). The integration of BG into the PV/WT system offers substantial cost-benefits by reducing the dependency on FC-based storage.
- The PV/WT/BG/FC generates the least GHG emissions of the three HRESs, and subsequently, the estimated effect on human health and ecosystem damage is also the lowest. In the yearly total 56,345 kg of CO_2 emissions from the PV/WT/BG/FC system, about 54% is from BG and the least (3%) is caused by the WT. In the process of implementing the optimal HRES, the total kerosene consumption savings would be around 30,950 l/yr and it would likely mitigate the emission of around 75,832 kg/yr of CO_2.
- The annual carbon penalty cost due to GHG emissions is found to be $845 for the PV/WT/BG/FC-based ROD system, whereas the costs for the PV/BG/FC and PV/WT/FC-based ROD systems are $906 and $1359, respectively.
- The HRES system with the desalination unit has substantial social benefits, including creating jobs and improving the human development index. Results from this analysis have confirmed estimated JC and HDI values remaining within 2.484–4.871 and 0.5877–0.6153, respectively.

In a world experiencing a transition in the energy sector, the opportunity to design energy systems capable of promoting sustainable socio-environmental development alongside revenue generation is enormous. While the present study takes into account some of the crucial design factors to model the proposed system, there are emerging issues regarding renewable micro-grids that require attention. For instance, the safety and security of hybrid components (from theft) can be an issue with the implementation of these hybrid systems in remote places. In some areas, the overall understanding of the electrification system in the community can create a dissonance between the objective of the modelers and common people: the consumers, even the stakeholders. Therefore, investigation of social acceptance can be of profound importance. The future works should consider these matters. Moreover, the future works should also investigate the resilience of the stand-alone microgrids during natural disasters.

CRediT authorship contribution statement

Dibyendu Roy: Conceptualization, Methodology, Investigation, Data curation, Writing – original draft, Writing – review & editing. **Rakibul Hassan:** Conceptualization, Methodology, Software, Visualization, Writing – original draft, Writing – review & editing. **Barun K. Das:** Conceptualization, Methodology, Software, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix

Table A1
Technical and economic details of the proposed HRES.

Components	Technical specifications	Data	Economic details	Data
PV module (Cano et al., 2020SunPower, 2016)				
	Model	SPR-E20-327	Capital cost	1300$/kW
	Nominal power, R_{PV}	327W	O&M cost	20 $/kW/yr
	Panel efficiency, η_{PV}	21.4%	Replacement cost	0
	Rated voltage, V_{mpp}	54.7V		
	Rated current, I_{mpp}	5.98A		
	Power temp. coefficient, α_p	−0.35%/°C		
	Derating factor, f_{pv}	88%		
	Lifetime	25yrs		
Wind Turbine (Akhtari et al., 2020)	Rated capacity, R_{WT}	10 kW	Capital cost	1295 $/kW
	Rotor diameter, d	15.81 m	O&M cost	25.2 $/kW/yr
	Cut-in wind speed, $V_{cut\text{-}in}$	2.75 m/s	Replacement cost	0
	Rated wind speed, V_{rated}	6.5 m/s		
	Cut-out wind speed, $V_{cut\text{-}out}$	20 m/s		
	Hub height, H	20 m		
	Lifetime	20 years		
Biomass gasifier (Rajbongshi et al., 2017)				
	Rated capacity	50 kW	Capital cost	1600$/kW
	Lifetime	25,000 h	O&M cost	0.025$/h
			Replacement cost	1280$/kW
Inverter (Das et al., 2021c)	Nominal power, P_{inv}	1 kW	Capital cost	300 $/kW
	Conversion efficiency, η_{inv}	96%	O&M cost	0
	Lifetime	15 years	Replacement cost	300 $/kW

(continued on next page)

Table A1 (*continued*)

Components	Technical specifications	Data	Economic details	Data
Fuel cell (Jamshidi and Askarzadeh, 2019)				
	Rated power	2 kW	Capital cost	3000 $/kW
	FC efficiency, η_{fc}	50%	O&M cost	0.02 x Capital cost
	Lifetime	5 years	Replacement cost	2500 $/kW
H_2 tank (Jamshidi and Askarzadeh, 2019)				
	Rated capacity	6 kg	Capital cost	660 $/kW
	Lifetime	20 years	O&M cost	0.02 x Capital cost
			Replacement cost	660 $/kW
Electrolyzer (Jamshidi and Askarzadeh, 2019)				
	Rated capacity	2 kW	Capital cost	2000 $/kW
	Lifetime	15 years	O&M cost	20 $/unit
			Replacement cost	1500 $/kW
RO system (Wu et al., 2018)				
	Rated capacity	30 m^3/day	Capital cost	532 $/$m^3$/day
	Annual number of membrane	2	O&M cost	0.2 $/$m^3$
			Replacement cost	0.06 $/$m^3$
			Chemical cost	0.06 $/$m^3$
			Water tank cost	255.4 $/$m^3$

Nomenclature

A_{sw}	Total swept area (m^2)
C_{cap}	Capital cost ($)
C_{fuel}	Fuel cost ($)
$C_{O\&M}$	Operation and maintenance cost ($)
C_{rep}	Replacement cost ($)
C_{sal}	Salvage value($)
E	Energy generation (kWh)
f_{PV}	Derating factor of PV (%)
F_{fuel}	Annual fuel consumed(L/year)
H	Hours of operation (hours)
i	Annual interest rate (%)
I	Solar irradiation (kW/m^2)
L	Project lifetime (year)
N_{comp}	Number of components
P_{fuel}	Fuel cost ($/L)
T_c	PV cell temperature (̊C)
T_s	Cell temperature at standard conditions (̊C)
$V_{cut.in}$	Cut-in wind velocity(m/s)
$V_{cut.off}$	Cut-off wind velocity(m/s)
W_{PV}	Power generation by PV (kW)
W_{rated}	Rated power (kW)
W_{WT}	Power generation by WT(kW)

Greek symbols

μ	Temperature coefficient
η_{BG}	Conversion efficiency of BG(%)
η_{inv}	Inverter efficiency (%)
η_{PV}	PV efficiency (%)
η_w	WT efficiency (%)
ψ	lifetime equivalent CO_2 emissions

Abbreviations

BAT	Battery
BG	Biomass generator
CF	Capital utilization factor
CRF	Capital recovery factor
COE	Cost of energy
DG	Diesel generator
FC	Fuel cell
HDI	Human development index
HHD	human health damage
HHV	Higher heating value

HRES	Hybrid renewable energy system
INV	Inverter
JC	Job creation
LCE	Life cycle emission
LCOE	Levelized cost of electricity
LPSP	Loss of power supply probability
NSGA-II	Non-dominated sorting genetic algorithm II
NPC	Net present cost
PV	Photovoltaic
ROD	Reverse osmosis desalination
WT	Wind turbine

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